i.e.,

$(\Lambda^{\circ}_m)_{KCl} - (\Lambda^{\circ}_m)_{NaCl} = 149.86 - 126.45$

$\{(\lambda^{\circ}_m)_{K^+} + (\lambda^{\circ}_m)_{Cl^-}\} - \{(\lambda^{\circ}_m)_{Na^+} + (\lambda^{\circ}_m)_{Cl^-}\} = 23.41$

$(\lambda^{\circ}_m)_{K^+} - (\lambda^{\circ}_m)_{Na^+} = 23.41$

Similarly, we can conclude that $(\lambda^{\circ}_m)_{Br^-} - (\lambda^{\circ}_m)_{Cl^-} = 2.06$

Electrolyte	Λ^0_m at 298 K	Difference
KCl	149.86	
NaCl	126.45	23.41
KBr	151.92	
NaBr	128.51	23.41
KNO_3	114.96	
$NaNO_3$	121.55	23.41

Electrolyte	Λ^0_m at 298 K	Difference
KBr	151.92	
KCl	149.86	2.06
NaBr	128.51	
NaCl	126.45	2.06
LiBr	117.09	
LiCl	115.03	2.06

Applications of Kohlrausch's Law

1. Calculation of molar conductance at infinite dilution of a weak electrolyte.

It is impossible to determine the molar conductance at infinite dilution for weak electrolytes experimentally. However, the same can be calculated using Kohlraush's Law.

For example, the molar conductance of CH_3COOH, can be calculated using the experimentally determined molar conductivities of strong electrolytes HCl, NaCl and CH_3COONa .

$$\Lambda^{\circ}_{CH_3COONa} = \lambda^{\circ}_{Na^+} + \lambda^{\circ}_{CH_3COO^-} \quad .....(1)$$

$$\Lambda^{\circ}_{HCl} = \lambda^{\circ}_{H^+} + \lambda^{\circ}_{Cl^-} \quad .....(2)$$

$$\Lambda^{\circ}_{NaCl} = \lambda^{\circ}_{Na^+} + \lambda^{\circ}_{Cl^-} \quad .....(3)$$

Equation (1) + Equation (2) – Equation (3) gives,

$$\left(\Lambda^{\circ}_{CH_3COONa}\right) + \left(\Lambda^{\circ}_{HCl}\right) - \left(\Lambda^{\circ}_{NaCl}\right) = \lambda^{\circ}_{H^+} + \lambda^{\circ}_{CH_3COO^-}$$

$$= \Lambda^{\circ}_{CH_3COOH}$$

2. Calculation of degree of dissociation of weak electrolytes

The degree of dissociation of weak electrolyte can be calculated from the molar conductivity at a given concentration and the molar conductivity at infinite dilution using the following expression

$$\alpha = \frac{\Lambda_m}{\Lambda^{\circ}_m} \quad .....(9.14)$$

Calculation of dissociation constant using Λ_m values. According to Ostwald dilution Law,

$$K_a = \frac{\alpha^2 C}{(1-\alpha)} \quad .....(9.15)$$

Substitute α value in the above expression (9.15)

$$K_a = \frac{\Lambda^2_m \, C}{\Lambda^{o\,2}_m \left(1 - \frac{\Lambda_m}{\Lambda^{\circ}_m}\right)}$$

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